Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Niobium crystallizes in body centred cubic structure. If density is $8.55\ \mathrm{g\ cm^{-3}}$, calculate atomic radius of niobium. $(\mathrm{M_w}$ of niobium $= 93)$

(A) 1.43×10^{-8} cm.

(B) 2.43×10^{-8} cm.

(C) 0.43×10^{-8} cm.

(D) 1.43×10^{-4} cm.

Q.2

The number of atoms in $100\ \mathrm{g}$ of a bcc crystal lattice with density $= 10\ \mathrm{g/cc}$ and edge length $= 100\mathrm{pm}$ is

(A) 4×10^{25}

(B) 2×10^{25}

(C) 8×10^{25}

(D) 1×10^{25}

Q.3

If the unit cell of a mineral has a cubic close-packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions, m and n, respectively, are:

(A) $1/2, 1/8$

(B) $1, 1/4$

(C) $1/2, 1/2$

(D) $1/4, 1/8$

Q.4

The percentage void space per unit volume of unit cell in zinc fluoride structure is -

(A) 15.03%

(B) 22.18%

(C) 18.23%

(D) 25.07%

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Q.5

FeO, crystal has a cubical structure and each edge of the unit cell is $5\ A$. Taking density of the oxide as $4gm/cc$, the number of Fe^{2+} and O^{2-} ions present in each unit cell are

(A) $4Fe^{2+}$ and $4O^{2-}$

(B) $6Fe^{2+}$ and $8O^{2-}$

(C) $2Fe^{2+}$ and $2.5O^{2-}$

(D) $1Fe^{2+}$ and $1.5O^{2-}$

Q.6

The olivine series of minerals consists of crystals in which Fe and Mg ions may substitute for each other causing substitutional impurity defect without changing the volume of the unit cell. In olivine series of minerals, oxide ion exist as FCC with Si^{4+} occupying $1/4^{th}$ of octahedral voids and divalent ions occupying $1/4$ th of tetrahedral voids. The density of forsterite (magnesium silicate) is $3.21\ g/cc$ and that of fayalite (ferrous silicate) is $4.34\ g/cc$. Find the percentage of fayalite in an olivine with a density of $3.88\ g/cc$.

(A) 59.29

(B) 39.29

(C) 69.29

(D) 72.38

Q.7

A compound formed by elements X and Y, crystallizes in the cubic structure, where X is at the corners of the cube and Y is at the six face centres. If side length is $5\mathring{A}$, estimate the density of the solid assuming atomic weight of X and Y as 60 and 90 respectively.

(A) $3.21gm/cm^2$

(B) $4.38gm/cm^2$

(C) $5.21gm/cm^2$

(D) $6.21gm/cm^2$

Q.8

A crystalline solid is made of X, Y and Z elements, Atoms of X form FCC packing: atoms of Y occupy octahedral voids while atoms of Z occupy tetrahedral voids. What will be the simplest formula of solid if atoms along one body diagonal are removed

(A) $X_5Y_4Z_8$

(B) XYZ

(C) $X_8Y_4Z_5$

(D) X_3YZ

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Q.9

$CsCl$ has bcc structure. If atomic mass of Cs and Cl atom is 133 and 35.5 amu respectively, what would be correct about its unit cell?

(A) Number of formula unit $(z) = 1$

(B) Number of formula unit $(z) = 2$

(C) Mass of unit cell $= 370 \text{ g mol}^{-1}$

(D) Mass of unit cell $= 168.5 \text{ g mol}^{-1}$

Q.10

Which of the following is/are incorrect for Rock salt crystal-

(A) If all Na^+ ions occupied at body centre are removed according to schottky defect then formula of resulting ionic compound would be Na_3Cl_4.

(B) If Na^+ ions would be present in alternate tetrahedral voids then formula would be Na_2Cl.

(C) If some Cl^- get displaced from the lattice in the presence of Na vapours then colour of compound will change.

(D) If interionic distance in crystal is $4.2\mathring{A}$
then edge length of unit cell becomes equal to $8.4\mathring{A}$

Q.11

If in $NaCl$ crystal, two 'F-centres' are present per unit cell then the possible formula per unit cell is/are

(A) $Na_4Cl_3.75$

(B) Na_4Cl_3

(C) $NaCl$

(D) $Na_4Cl_3.375$

Q.12

Paragraph:
Potassium crystallizes in a body-centred cubic lattice with edge length $a = 5.2\mathring{A}$.
Question:
What is the distance between nearest neighbours is -

(A) $4.5\mathring{A}$

(B) $5.2\mathring{A}$

(C) $3.2\mathring{A}$

(D) $6.4\mathring{A}$

Q.13

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Paragraph:

Potassium crystallizes in a body-centred cubic lattice with edge length $\mathrm{a} = 5.2\mathring{A}$.

Question:

What is the distance between next-nearest neighbours?

(A) $4.5\mathring{A}$

(B) $5.2\mathring{A}$

(C) $3.2\mathring{A}$

(D) $6.4\mathring{A}$

Q.14

Paragraph:

Potassium crystallizes in a body-centred cubic lattice with edge length $\mathrm{a} = 5.2\mathring{A}$.

Question:

How many nearest neigbours does each K atom have ?

(A) 2

(B) 4

(C) 6

(D) 8

Q.15

Paragraph:

A gold single crystal has a cubic shape and the dimension of the cube is a $= 1.000\ \mathrm{cm}$. When irradiated with X-rays $(\lambda = 154.05\mathrm{pm})$ at the angle θ of 10.89° it gives a well-defined first-order diffraction pattern. The atomic weight of Au is $\mathrm{M_{Au}} = 196.97\ \mathrm{g\ mol^{-1}}$.

Question:

How many gold atoms are in the cube ?

(A) 5.904×10^{22}

(B) 5.904×10^{20}

(C) 5.904×10^{18}

(D) 5.904×10^{16}

Q.16

Paragraph:

A gold single crystal has a cubic shape and the dimension of the cube is a $= 1.000\ \mathrm{cm}$. When irradiated with X-rays $(\lambda = 154.05\mathrm{pm})$ at the angle θ of 10.89° it gives a well-defined first-order diffraction pattern. The atomic weight of Au is $\mathrm{M_{Au}} = 196.97\ \mathrm{g\ mol^{-1}}$.

Question:

What is the mass of the unit cell of gold ?

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(A) 1.308×10^{-19}

(B) 1.308×10^{-21}

(C) 1.308×10^{-17}

(D) 1.308×10^{-15}

Q.17

Paragraph:

A gold single crystal has a cubic shape and the dimension of the cube is a $= 1.000\ \text{cm}$. When irradiated with X-rays $(\lambda = 154.05\text{pm})$ at the angle θ of 10.89° it gives a well-defined first-order diffraction pattern. The atomic weight of Au is $M_{Au} = 196.97\ \text{g mol}^{-1}$.

Question:

What is the density of gold?

(A) $29.31\ \text{g/cm}^3$

(B) $26.29\ \text{g/cm}^3$

(C) $14.21\ \text{g/cm}^3$

(D) $19.31\ \text{g/cm}^3$

Q.18

Match the column:

Column I	**Column II**
(A) Rock salt structure	(p) Coordination number of cation is 4
(B) Zinc blend structure	(q) $\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C) Flourite structure	(r) Coordination number of cation and anion are same
(D) Structure of Cesium halide	(s) Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

(A) A-r,s; B-p,q,r,s ; C- q ; D-r;

(B) A-p,q,r,; B-r ; C- s ; D-s;

(C) A-s; B-r,p,q ; C-s ; D-s;

(D) A-s,r; B-p ; C- p ; D-s;

Q.19

A cubical shaped crystal of a compound $\text{A}_2\ \text{B}$ (antiflourite structure (Na_2O type)) having volume 64 and density of compound is found to be 50gm/cm^3.

If number of cations present in body diagonal plane of cubical shaped crystal $= X$ then value of $(\text{X} - 8)$ is :

(Given that : formula mass of $\text{A}_2\ \text{B} = 60.23\text{amu}$ and $N_A = 6.023 \times 10^{23}$)

Q.20

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A compound AB has zinc blende structure with 'A' atoms at corners and face centre of the unit cell. If all the atoms present in one body diagonal are removed, the formula of the compound becomes A_xB_y. The sum $(x+y)$ is:

Answers & Solutions

Q.1 Answer:

—

Solution:

Given that Crystal density $= 8.55\ \text{g cm}^{-3}$.

For $bcc, n = 2$

Mw of niobium $= 93$

$\because$ **Crystal density** $= \frac{\text{n} \times \text{M}_\text{w}}{\text{V} \times \text{N}_\text{A}}$

$\therefore \quad \text{V} = \frac{2 \times 93}{8.55 \times 6.023 \times 10^{23}} = 3.6 \times 10^{-23}\ \text{cm}^3 \quad$ **or** $\quad \text{a}^3 = 36 \times 10^{-24}\ \text{cm}^3$

$\therefore \quad \text{a} = 3.3 \times 10^{-24}\ \text{cm}$

For bcc, $4r = \sqrt{3}a$

$\therefore \quad \text{r} = \frac{1.732 \times 3.3 \times 10^{-8}}{4} = 1.43 \times 10^{-8}\ \text{cm.}$

Q.2 Answer:

2×10^{25}

Solution:

For a BCC unit cell, $Z = 2$

$\rho = 10\ \text{g/cc} : \text{a} = 100 \times 10^{-12} \times 10^2\ \text{cm}$

$\therefore \text{a} = 1 \times 10^{-8}\ \text{cm}$

$$\rho = \frac{\text{Z} \times \text{M}_\text{w}}{\text{a}^3 \times \text{N}_\text{A}}$$

$$\text{M}_\text{w} = \frac{10 \times \left(10^{-8}\right)^3 \times 6.023 \times 10^{23}}{2} = 10 \times 10^{-24} \times 10^{23} \times \frac{6.023}{2} = 3.016$$

$3.016\ \text{g} \Leftrightarrow$ **Avogadro number of atoms,** N_A

$$1\ \text{g} \Leftrightarrow \frac{\text{N}_\text{A}}{3.016}$$

$$100\ \text{g} \Leftrightarrow \frac{\text{N}_\text{A}}{3.016} \times 100 = \frac{6.023 \times 10^{23} \times 100}{3.016}$$

$\therefore \quad$ **Number of atoms** $= 2 \times 10^{25}$

Q.3 Answer:

$1/2, 1/8$

Solution:

In ccp array:

Number of oxygen atoms $= 4$

Number of octahedral holes $= 4$

Number of tetrahedral holes $= 8$

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Number of Al^{3+} ions $= 4 \times m$

Number of Mg^{2+} ions $= 8 \times m$

Molecule of mineral will be neutral, hence

$4(-2) + 4m(+3) + 8n(+2) = 0$

$12m + 16n = 8$

$3m + 4n = 2$

$\therefore m = 1/2$ and $n = 1/8$

Q.4 Answer:

25.07%

Solution:

Anions occupy fcc positions and half of the tetrahedral holes are occupied by cations. Since there are four anions and 8 tetrahedral holes per unit cell, the fraction of volume occupied by spheres per unit volume of the unit cell is

$= \frac{4\times\left(\frac{4}{3}\pi r_a^3\right)+\frac{1}{2}\times 8\times\left(\frac{4}{3}\pi r_c^3\right)}{16\sqrt{2}r_a^3} = \frac{\pi}{3\sqrt{2}}\left\{1+\left(\frac{r_c}{r_a}\right)^3\right\}$

For tetrahedral holes, $\frac{r_c}{r_a} = 0.225 = \frac{\pi}{3\sqrt{2}}\left\{1+(0.225)^3\right\} = 0.7493$

Void volume $= 1 - 0.7493 = 0.2507/$ **unit volume of unit cell .**

$\%$ **void space** $= 25.07\%$

Q.5 Answer:

$4Fe^{2+}$ and $4O^{2-}$

Solution:

Volume of unit cell $= \left(5 \times 10^{-8}\right)^3$ cc $= 1.25 \times 10^{-22}$cc

Density of $FeO = 4gm/cc$

Mass of unit cell $= 1.25 \times 10^{-22} \times 4 = 5 \times 10^{-22}$gm

Mass of $FeO = 72gm/mole$, **For 1 molecule** $= \frac{72}{6.022\times10^{23}}$

Hence number of FeO **molecules per unit cell** $= \frac{5\times10^{-22}}{1.195\times10^{-22}}$

Hence there are four Fe^{2+} **and four** O^{2-} **ions in each unit cell**

Q.6 Answer:

59.29

Solution:

According to given data:

Number of atoms of $O^{2-} = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

Number of atoms of $Si^{i^{4+}} = \frac{1}{4}\times$ **octahedral void** $= \frac{1}{4} \times 4 = 1$

Number of atoms of $Mg^{2+} = \frac{1}{4}\times$ **tetrahedral void** $= \frac{1}{4} \times 8 = 2$

$\because$ **Forsterite** $= Mg_2SiO_4$ **and Fayalite** $= Fe_2SiO_4$

Let the forsterite is $x\%$ **and fayalite is** $(100 - x)\%$, **then** $\frac{x\times3.21+(100-x)\times4.34}{100} = 3.88 \Rightarrow x = 40.71\%$

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(Forsterite)

Therefore, percentage of Fayalite is 59.29%.

Q.7 Answer:

$4.38\mathrm{gm/cm^2}$

Solution:

From eight corner atoms, one atom (X) contributes to one unit cell. From six face centres, three atoms (Y) contribute to one unit cell.

So, the formula of the compound is $\mathrm{XY_3}$.

As we know that, $\rho = \frac{\mathrm{Z \times M_m}}{\mathrm{N_A \times a^3}}$, here $\mathrm{Z} = 1$

Molar mass of $\mathrm{XY_3}$

$\mathrm{M_m} = 60 + 3 \times 90 = 330\mathrm{gm}$

$\rho = \frac{1 \times 330}{6.023 \times 10^{23} \times (5 \times 10^{-8})^3} = \mathrm{gm/cm^3}$

$= \frac{330}{6.023 \times 10^{23} \times 125 \times 10^{-24}} \mathrm{gm/cm^3} = 4.38\mathrm{gm/cm^2}$

For octahedral void, $\mathrm{R} = 0.414 \times \frac{362}{2\sqrt{2}} = 53\mathrm{pm}$

Q.8 Answer:

$\mathrm{X_5Y_4Z_8}$

Solution:

Number of atoms of X (at packing sile, t.e., at corners and face-centres)

$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

Number of atoms of $\mathrm{Y} = 4$

Number of atoms of $\mathrm{Z} = 8$

Along one body diagonal there will be two X atoms, one Y atom and two Z atoms are found and are removed.

Number of atoms of X will be $= 4 - \frac{1}{8} \times 2 = \frac{15}{4}$

Number of atoms of Y will be $= 4 - 1 = 3$

Number of atoms of 7 will be $= 8 - 2 = 6$

$X : Y : X$

$\frac{15}{4} : 3 : 6$

$5 : 4 : 8$

$\therefore$ **Simplest formula will be** $X_5Y_4Z_8$

Q.9 Answer:

Number of formula unit $(z) = 1$

Solution:

CsCl **has** BCC **structure**

$\mathrm{Cs} = 133, \mathrm{Cl} = 35.5$

$\mathrm{Cs^+}$ **is present at the centre and** $8\mathrm{Cl^-}$ **ions are present at the corners.**

$\therefore$ **No. of formula unit** $(z) = 1$

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Mass of unit cell $= 133 + 35.5 = 168.5$

$\Rightarrow M_w = 133 + 35.5 = 168.5$

Q.10 Answer:

If all Na^+ ions occupied at body centre are removed according to schottky defect then formula of resulting ionic compound would be $\mathrm{Na_3Cl_4}$.

Solution:

In Schottky defect equal no. of cation and anion removes and formula would remain same. If Na^+ **would present in tetrahedral void then as** NaCl **formula must remain same.**

Due to F-centre colour of compound changes

$\mathrm{a} = 2\left(\mathrm{r}^+ + \mathrm{r}^-\right) = 2\times$ **inter ionic distance** $= 8.4\mathring{A}$

Q.11 Answer:

$\mathrm{Na_4Cl_3}.75$

Solution:

In F-centre, Cl^- **ions are removed.**

IstCombination : Two corner Cl^- **ions are removed.**

Number of Na^+ **ions** $= 4$

Number of Cl^- **ions** $= -3.75$

Hence, formula per unit cell is $\mathrm{Na_4Cl_3}.75$.

IInd Combination : Two face Cl^- **ions are removed**

Number of Na^+ **ions** $= 4$

Number of Cl^- **ions** $== 3$

Hence, formula per unit cell is $\mathrm{Na_4Cl_3}$

IIIrdCombination :One corner and one face Cl^- **ions are removed.**

Number of Na^+ **ions** $= 4$

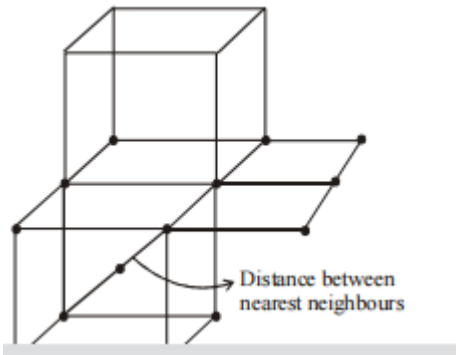
Number of Cl^- **ions** $== 3.375$

Hence, formula per unit cell is $\mathrm{Na_4Cl_{3.375}}$.

Q.12 Answer:

$4.5\mathring{A}$

Solution:



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(A) $a\sqrt{3} = 4r$

$\frac{a\sqrt{3}}{2} = 2r =$ **Distance between nearest neighbours.**

$$= 5.2 \times \frac{\sqrt{3}}{2} = 4.5\mathring{A}$$

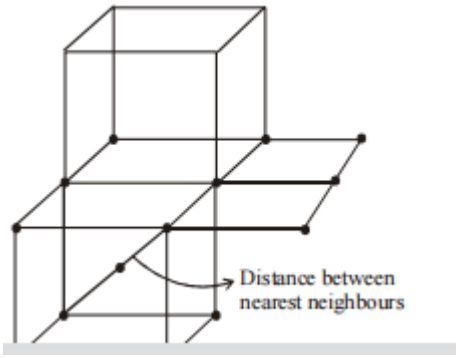
(B) Distance between next nearest neighbours $= a = 5.2\mathring{A}$

(C) Co-ordination number of K **atoms** $= 8$

Q.13 Answer:

$5.2\mathring{A}$

Solution:



(A) $a\sqrt{3} = 4r$

$\frac{a\sqrt{3}}{2} = 2r =$ **Distance between nearest neighbours.**

$$= 5.2 \times \frac{\sqrt{3}}{2} = 4.5\mathring{A}$$

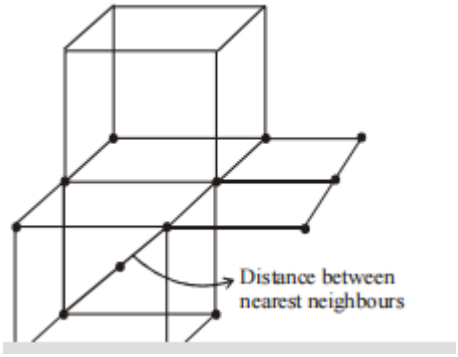
(B) Distance between next nearest neighbours $= a = 5.2\mathring{A}$

(C) Co-ordination number of K **atoms** $= 8$

Q.14 Answer:

8

Solution:



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(A) $a\sqrt{3} = 4r$

$\frac{a\sqrt{3}}{2} = 2r =$ **Distance between nearest neighbours.**

$$= 5.2 \times \frac{\sqrt{3}}{2} = 4.5\mathring{A}$$

(B) Distance between next nearest neighbours $= a = 5.2\mathring{A}$

(C) Co-ordination number of K **atoms** $= 8$

Q.15 Answer:

5.904×10^{22}

Solution:

Bragg law, $n\lambda = 2a \sin\theta$**, allows one to determine the lattice constant of Au according to the following:**

$a = \frac{n\lambda}{2\sin\theta} = \frac{1 \times 154.05 \times 10^{-12}}{2\sin(10.89^\circ)}$

$$a = 4.077 \times 10^{-10}\ m = 4.077 \times 10^{-8}\ cm$$

The volume of the crystallographic unit (unit cell) of Au equals:

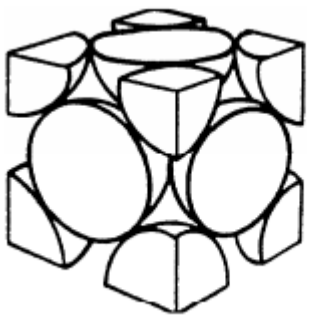
$$V_u = a^3 = \left(4.077 \times 10^{-10}\right)^3 = 6.777 \times 10^{-29}\ m^3 = 6.777 \times 10^{-23}\ cm^3$$

The number of crystallographic units of Au within $1.000\ cm^3$ **equals:**

$$N = \frac{1.000}{6.777 \times 10^{-23}} = 1.476 \times 10^{22}$$

Each crystallographic cell has four atoms, $n_u = 4$**. The corner atoms belong to eight unit cells, thus** $1/8$ **of each corner atom belongs to the cell; the face atoms belong to two unit cells, thus** $1/2$ **of each face atom belongs to the cell.**

$$n_u = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$



Face–centered cubic crystal structure

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The number of Au atoms in the $1.000\ \mathrm{cm}^3$ cube equals:

$$\mathrm{N_{Au}} = \mathrm{N} \times \mathrm{n_u} = 1.476 \times 10^{22} \times 4 = 5.904 \times 10^{22}$$

(b) The weight of one Au atom $(\mathrm{m_{Au}})$ equals:

$$\mathrm{m_{Au}} = \frac{\mathrm{M_{Au}}}{\mathrm{N_A}} = \frac{196.97}{6.002214 \times 10^{23}} = 3.271 \times 10^{-22}\ \mathrm{g}$$

The mass of the unit cell equals:
$\mathrm{M_u} = \mathrm{n_u m_{Au}} = 4 \times 3.271 \times 10^{-22} = 1.308 \times 10^{-21}\ \mathrm{g}$ **(c) The density of Au, thus the weight of the $1.000\ \mathrm{cm}^3$ cube, equals the number of unit cells within $1.000\ \mathrm{cm}^3$ times the mass of the cell:**

$$\mathrm{d_{Au}} = \mathrm{NM_u} = 1.476 \times 10^{22} \times 1.308 \times 10^{-21} = 19.31\ \mathrm{g\ cm^{-3}}$$

Q.16 Answer:

1.308×10^{-21}

Solution:

Bragg law, $\mathrm{n}\lambda = 2\mathrm{a}\sin\theta$, allows one to determine the lattice constant of Au according to the following:

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$$\mathrm{a} = 4.077 \times 10^{-10}\ \mathrm{m} = 4.077 \times 10^{-8}\ \mathrm{cm}$$

The volume of the crystallographic unit (unit cell) of Au equals:

$$\mathrm{V_u} = \mathrm{a}^3 = \left(4.077 \times 10^{-10}\right)^3 = 6.777 \times 10^{-29}\ \mathrm{m}^3 = 6.777 \times 10^{-23}\ \mathrm{cm}^3$$

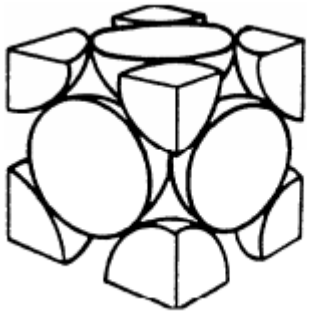
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$$\mathrm{n_u} = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

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Face–centered cubic crystal structure

The number of Au atoms in the $1.000\ \mathrm{cm}^3$ cube equals:

$$\mathrm{N_{Au}} = \mathrm{N} \times \mathrm{n_u} = 1.476 \times 10^{22} \times 4 = 5.904 \times 10^{22}$$

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Q.17 Answer:

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Solution:

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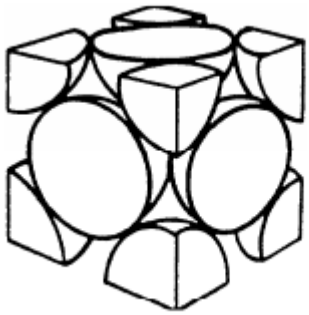
The number of crystallographic units of Au within $1.000\ \mathrm{cm}^3$ equals:

$$\mathrm{N} = \frac{1.000}{6.777 \times 10^{-23}} = 1.476 \times 10^{22}$$

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Each crystallographic cell has four atoms, $n_u = 4$. The corner atoms belong to eight unit cells, thus $1/8$ of each corner atom belongs to the cell; the face atoms belong to two unit cells, thus $1/2$ of each face atom belongs to the cell.

$$n_u = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$



Face–centered cubic crystal structure

The number of Au atoms in the $1.000\ cm^3$ cube equals:

$$N_{Au} = N \times n_u = 1.476 \times 10^{22} \times 4 = 5.904 \times 10^{22}$$

(b) The weight of one Au atom (m_{Au}) equals:

$$m_{Au} = \frac{M_{Au}}{N_A} = \frac{196.97}{6.002214 \times 10^{23}} = 3.271 \times 10^{-22}\ g$$

The mass of the unit cell equals:

$M_u = n_u m_{Au} = 4 \times 3.271 \times 10^{-22} = 1.308 \times 10^{-21}\ g$ (c) The density of Au, thus the weight of the $1.000\ cm^3$ cube, equals the number of unit cells within $1.000\ cm^3$ times the mass of the cell:

$$d_{Au} = NM_u = 1.476 \times 10^{22} \times 1.308 \times 10^{-21} = 19.31\ g\ cm^{-3}$$

Q.18 Answer:

A-r,s; B-p,q,r,s ; C- q ; D-r;

Solution:

((A) - rs, (B) - pqrs, (C) - q, (D) - r)

Rock-salt structure (AB type) : Chloride ions are considered as having cubic close packed structure. Sodium ions occupy octahedral holes. There are six octahedral holes around each chloride ion. The co-ordination no. of $Na^+ \& Cl^-$ is $6:6$.

The ionic radii ratio $= 0 \cdot 414$ (actual $= 0 \cdot 525$).

The no. of $NaCl$ units per unit cell $= 4$.

$r_c + r_a = \frac{a}{2}$ where r_c is the radius of Na^+ ion, r_a is the radius of Cl^- ion and ' a ' is the edge length of

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unit cell.

Zinc Blende Structure (Zinc Sulphide) : (AB type)

Sulphide ions have ccp lattice and zinc ions occupy half or total of tetrahedral voids.

Co-ordination No. is $4:4$. Radius ratio is 0.40 (actual $\cdot 225$). 4 ions per unit cell.

Structure of Calcium fluoride or fluorite : (AB $_2$ type)

Ca^{+2} ion have FCC lattice and Fluoride ions occupy all the tetrahedral voids in lattice. The co-ordination no. of calcium and fluoride ions in CaF_2 is $8:4$. No. of CaF_2 units per cell is 4 . $r_c + r_a = \frac{\sqrt{3}a}{4}$, where r_c is the radius of Ca^{2+} ion, r_a is the radius of F^-ion and ' a ' is the edge length of unit cell.

Structure of Cesium halide : This type of structure is exhibited by cesium chloride. Chloride ions exist as primitive cubic and cesium ions occupy centered position (cubic void). Overall this structure is referred as BCC-like structure. The effective number of $Cs^+$and Cl^- ions are one each, so the general formula is $CsCl$.

The co-ordination number of both the ions $\left(Cs^+ \text{ and } Cl^-\right)$is 8 , so the general formula using the coordination number of ions is Cs_8Cl_8 or $CsCl$. This kind of structure is also exhibited by other halides of cesium and by the halides of ammonium like NH_4Cl, NH_4Br, etc.

Q.19 Answer:

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Solution:

$\rho = \frac{Z \times M}{N_A \times a^3}$ **where** $Z =$ **no. of formula units per unit cell and** $a^3 =$ **volume of unit-cell** $50 = \frac{4 \times 60.23}{6.023 \times 10^{23} \times a^3} \Rightarrow a^3 = 8 \times 10^{-24}\ cm^3 = 8$ **volume of cubical shaped crystal = no. of unit cells** $\times a^3$

$\Rightarrow 64 =$ **no. of unit cells** $\times 8$

$\Rightarrow$ **number of unit cells** $= 8$

therefore along one body diagonal plane 4 unit cells are present and number of cations present in body diagonal plane of one unit cell is 4 therefore total no. of cations $= 16$

Q.20 Answer:

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Solution:

A present in corners and face centre.

B present in alternate tetrahedral voids.

No. of A **per unit cell** $= 6 \times \frac{1}{8} + 6 \times \frac{1}{2} = \frac{15}{4}$

No. of B **per unit cell** $= 4 - 1 = 3$

$$x : y = \frac{15}{4} : 3 = 5 : 4$$

$$x + y = 5 + 4 = 9$$

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